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# WEKA Experiment

| S.No | Date    | Title                                                             | Sign |
|------|---------|-------------------------------------------------------------------|------|
| 1.   | 4/8/26  | data preprocessing and analysis for dataset                       |      |
| 2.   | 4/8/26  | Data segmentation by k-means cluster                              |      |
| 3.   | 4/8/26  | Data segmentation expectation maximization Algorithm.             |      |
| 4.   | 6/8/26  | Data segmentation By cloud-hierarchical clustering Algorithm      |      |
| 5.   | 6/8/26  | Frequent pattern mining using Association Rule                    |      |
| 6.   | 6/8/26  | Frequent pattern mining using Fp Growth                           |      |
| 7.   | 11/8/26 | prediction of categorical data using decision tree algorithm      |      |
| 8.   | 11/8/26 | prediction of categorical data using smo algorithm                |      |
| 9.   | 11/8/26 | Evaluating Accuracy of the classifiers                            |      |
| 10.  | 13/8/26 | Description Numerical prediction Analysis using linear Regression |      |

Completed  
Akash  
13/8/26



## Experimental - 01 : Addition

Aim :- To prove the program for addition using R-Tool

Program :-

```
num1 <- as.integer(readline(prompt = "Enter the  
first number: "))
```

```
num2 <- as.integer(readline(prompt = "Enter the  
second number: "))
```

```
num3 <- num1 + num2
```

```
print(num3)
```

Output :-

```
z = num1 + num2  
= 10 + 18  
= 28
```

Result :- Thus the basic program addition is executed successfully. ✓

## Experiment - 02 : Subtraction

Aim :- To prove program for subtraction using R-tool

Program :-

```
x → num1
```

```
y → num2
```

```
z → x - y
```

```
Print(z)
```

Output :-

```
z = num1 - num2  
= 10 - 8 = 2
```

Result :- Thus the basic program subtraction are executed successfully.

### Experiment -3      Multiplication

Aim:- To prove the program for multiplication using R-tool.

Program:-

```
x = num1  
y = num2  
z = num1 * num2  
print(z)
```

Output:-  $z = \text{num1} * \text{num2}$   
 $= 5 * 3 = 15$

Result:- Thus the basic program multiplication are executed successfully.

### Experiment -4      Division

Aim:- To prove the program for division using R-tool.

Program:-

```
x = num1  
y = num2  
z = num1 / num2  
print(z)
```

Output:-

$z = \text{num1} / \text{num2}$   
 $= 10 / 5$   
 $= 2$

Result:- Thus the basic program division was executed successfully.



## Experiment -05    odd (or) even

Aim:- To write the program for odd or even using R-Tool

Program:-

```
num = 45
if ((num % 2) == 0) {
  print(paste(num, "is even"))
} else {
  print(paste(num, "is odd"))
}
```

output:- "45 is odd"

Result:- Thus basic program odd (or) even executed successfully

## Experiment -06    Mean, median, mode

Aim:- To program for mean, median, mode

Program:-

```
x <- (2, 4, 8, 7, 4, 6, 4)
mean_value <- mean(x)
median_value <- median(x)
mode_function <- function(x)
unique_v <- unique(x)
unique_v [which.max(table(match(x, unique_v)))]
}
mode_value <- mode_function(x)
mean_value
median_value
mode_value
```

output:-

mean - value - 4.85714 = 5

mode - value - 5

median - value - 6

Result:-

Thus, the program for mean, median, mode has been successfully executed.

### experiment - 7 summary

Aim:- TO implement the program for summary using R-Tool

Program:-

```
age <- c(23, 24, 25)
```

```
summary(age)
```

```
write.csv(data.frame(age), "datafr.csv")
```

output:-

| min  | 1st Qu | median | mean | 3rd Qu | max  |
|------|--------|--------|------|--------|------|
| 23.0 | 23.5   | 24.0   | 24.0 | 24.5   | 25.0 |

Result:-

Thus, the program for summary has been executed successfully.



## Experiment-8 Greater among three numbers

Aim:- To write the program for greatest among three numbers

Program:-

x = 5

y = 8

z = 7

print(paste("greatest is :", max(x,y,z)))

Output:-

Greatest is : 8

Result:-

Thus, the program for greater among three numbers has been executed successfully

## Experiment-9 IQR

Aim:- To write the program for central tendency and data dispersion measures using R-tool

Program:-

x <- c(4,10,20,20)

IQR - value <- IQR(x)

IQR - value

Output:-

IQR - value = 5.5

Result

Thus the program for central tendency and data dispersion measures was executed successfully.

## Experiment - 10      Quantile

Aim:- To write the program for central tendency and data dispersion measures.

Program:-

$x \leftarrow (5, 10, 15, 20, 25)$

quantile(x)

Output:-

quantile(x)

0%    25%    50%    75%    100%

5        10        15        20        25

Result:-

Thus the program for central tendency and data dispersion measures was executed successfully.

## Experiment - 11      Mid Range

Aim:- To write the program for central tendency and data dispersion measures.

Program:-

$x \leftarrow (20, 30, 40, 50, 60)$

mid-range  $\leftarrow ((\min(x) + \max(x)) / 2)$

mid-range

Output:-

40

Result:-

Thus the program for central tendency and data dispersion measures was executed successfully.

## Experiment -12      z-score Normalization

Aim:- To write the program for z-score normalization using R-Tool.

Program:-

```
x <- c(
z-score <- (x - mean(x)) / sd(x)
z-score
```

Output:-

z-score

-0.6892090    0.00000    -0.6035829    1.7245225    -0.4311306

Result:- Thus the program has been successfully executed.

## Experiment -13      Min, Max, Mean, MinMax

Aim:- To write the program for minimum, maximum, mean and minimum using R-Tool.

Program:-

```
x <- c(
min-value <- min(x)
max-value <- max(x)
minmax - norm <- (x - min-value) / (max-val - min-val)
min-value
max-value
mean-value
minmax - norm.
```



output:-

min-value - 10

max-value - 50

mean-value - 23.2

minmax-norm - 0.00 0.05 0.50 0.10 1.00

Result:-

Thus the program has been successfully executed.

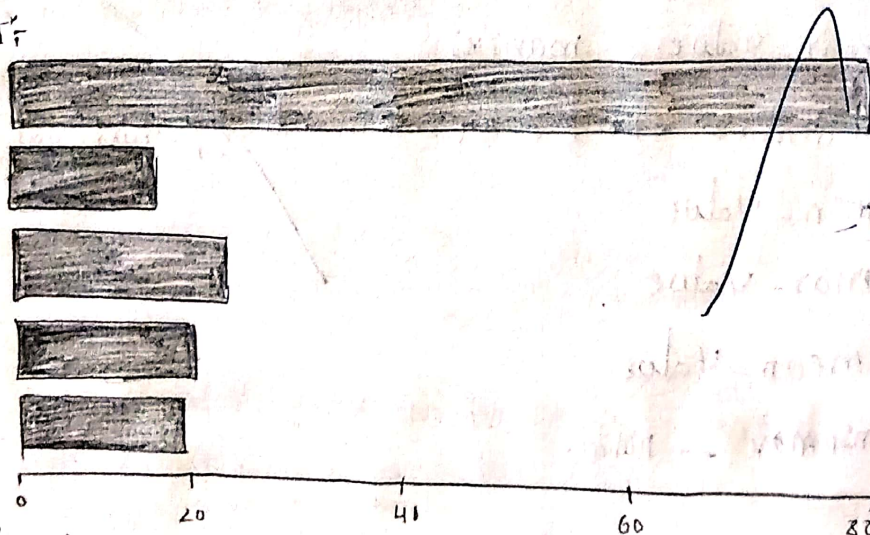
### Experiment - 14 Barplot and Horizontal Bar

Aim:- To draw the bar plot and horizontal bar using R-tool.

Program:-

```
a <- c(15, 17, 19, 10, 90)
barplot(a)
a <- c(15, 17, 19, 10, 90)
barplot(a)
barplot(a, horiz = TRUE)
```

output:-



Result:-

Thus the bar and horizontal barplot was executed successfully.

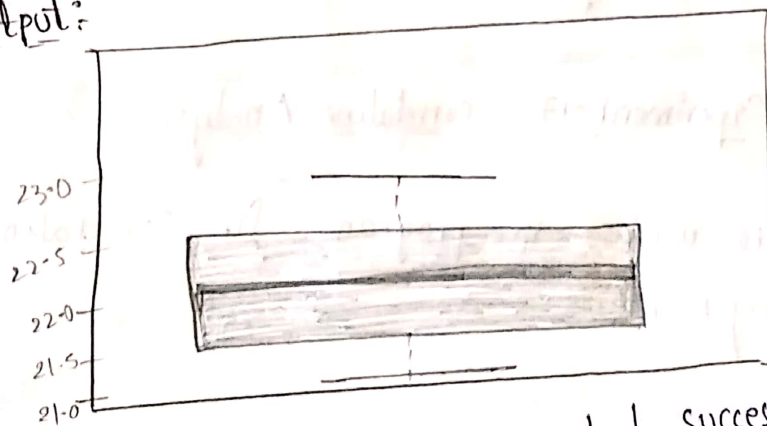
## Experiment -15      Box plot

Aim:- To draw the box plot using R-Tool

Program:-

```
name <- c("gita", "sita", "radha")  
age <- c(21, 22, 23)  
marks <- c(78, 86, 35)  
df <- data.frame(name, age, marks)  
boxplot(df$age)  
hist(df$age)
```

Output:-



Result:-  
Thus the box plot was executed successfully

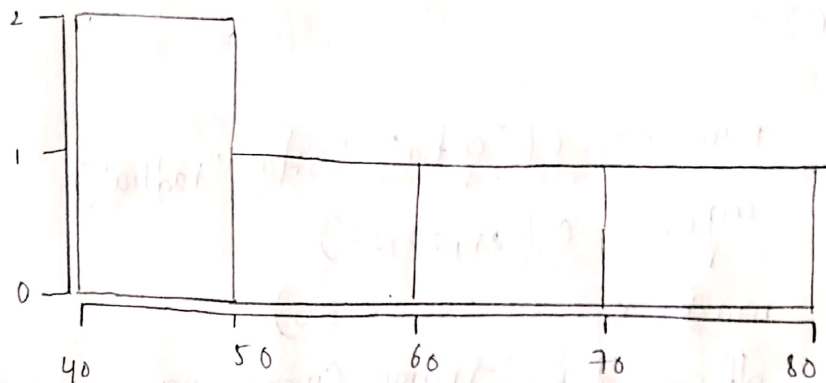
## Experiment -16      Histogram

Aim:- To draw the histogram plot using R-Tool

Program:-

```
a <- c(45, 47, 59, 70, 80)  
hist(a)
```

output:-



Result:-

Thus the histogram plot was executed successfully.

### Experiment-17 correlation Analysis

Aim:- To write the program for correlation analysis using R-Tool.

Program:-

```
x <- c(10, 20, 30, 40, 50)
```

```
y <- c(15, 25, 35, 45, 55)
```

```
correlation <- cor(x, y)
```

```
cat("correlation coefficient is ", correlation)
```

output:- 1

Result:-

Thus the correlation analysis was executed successfully.



## Experiment 18

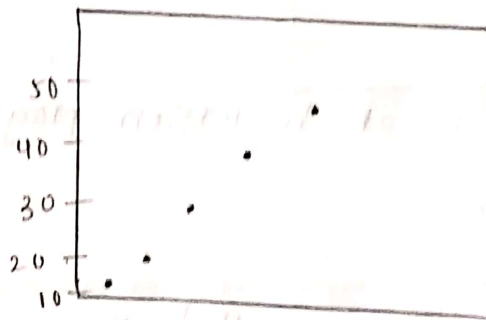
## Scatter plot

Aim:- To draw the scatter plot using R-tool

Program:-

```
set.seed(9)
x <- rnorm(1000)
y <- rnorm(1000)
smooth.scatter(y~x)
smooth.scatter(x~y)
smooth.scatter(x|y)
```

Output:-



Result:-

The program was executed successfully.

## Experiment 19 Linear Regression

Aim:- To write the program for linear regression using R-tool.

Program:-

```
x <- c(10, 20, 30, 40, 50)
y <- c(12, 15, 35, 45, 55)
model <- lm(y~x)
summary(model)
plot(x|y,
```

```
model = "linear regression"
```

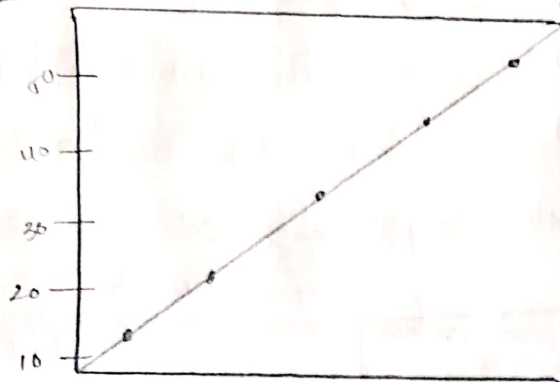
```
xlab = "x values"
```

```
ylab = "y values"
```

```
ch = 19
```

```
abline(model, col = "red")
```

output:-



Result:- Thus the linear regression program was executed successfully.

### experiment - 20      multiple Regression

Aim:- To write the program for the multiple regression.

Program:-

```
data <- dataframe (
```

```
  x1 <- c(10, 20, 30, 40, 50)
```

```
  x2 <- c(5, 19, 9, 11, 13)
```

```
  x  <- c(15, 25, 35, 45, 55)
```

```
  )
```

```
model <- lm(y ~ x1 + x2, data = data)
```

```
summary(model)
```

output:-

$\ln(\text{formula} = y \sim x_1 + x_2 \text{ data} = \text{data})$

| 1         | 2          | 3 | 4         | 5         |
|-----------|------------|---|-----------|-----------|
| 3.657e-16 | -2.481e-15 |   | 2.342e-15 | 1.297e-15 |

Result:-

The program was executed successfully using R-studio.

✓

as

e



# Heka experiment-1

## Apriori Algorithm

Aim:- To find frequent itemsets using Apriori algorithm.

Code:-

@ relation opriori

@ attribute Rice {yes, no}

@ attribute sugar {yes, no}

@ attribute Milk {yes, no}

@ attribute Tea {yes, no}

@ attribute Bread {yes, no}

@ attribute Butter {yes, no}

@ data

yes, yes, yes, yes, no, no, no

yes, no, yes, yes, yes, no, no

no, no, yes, no, yes, yes, no

yes, yes, no, yes, yes, no

output:-

1. Bread = no  $\Rightarrow$  Rice = yes

2. Rice = yes  $\Rightarrow$  Bread = no

3. Rice = yes  $\Rightarrow$  Butter = no

4. Sugar = yes  $\Rightarrow$  Butter = no

5. Bread = no  $\Rightarrow$  Butter = no

Result:-

Frequent patterns exist among Rice, sugar, milk.

## Experiment - 2

Aim: To generate frequent itemsets using Apriori  
→ Fp - Growth

Code:-

@ relation apriori 2

@ attribute B {yes, no}

@ attribute R {yes, no}

@ attribute E {yes, no}

@ attribute A {yes, no}

@ attribute D {yes, no}

@ attribute M {yes, no}

@ attribute U {yes, no}

@ attribute T {yes, no}

@ data

yes, yes, yes, yes, yes, yes, no

yes, yes, yes, yes, yes, yes, no

yes, no, no, no, no, yes, yes

yes, yes, yes, yes, yes, yes, yes

yes, yes, yes, yes, yes, yes, no

yes, no, no, no, no, yes, no

output:-

Apriori

Fp Growth

1.  $I = \text{no} \Rightarrow B = \text{yes}$

2.  $I = \text{no} \Rightarrow B = \text{yes}$

3.  $K = \text{no} \Rightarrow B = \text{yes}$

!

10.  $I = \text{no} \wedge L = \text{no} \Rightarrow S B = \text{yes}$

1.  $[I = \text{no}] : S \Rightarrow [K = \text{no}] : S$

$[K = \text{no}] : S \Rightarrow [L = \text{no}] : S$

$[I = \text{no}] : S \Rightarrow [I = \text{no}] : S$

!

$[I = \text{no}] : S \Rightarrow [I = \text{no}, K = \text{no}] : S$

Result:-

Fp-growth gives same result but faster

## Experiment-3

Aim:- To classify data using Decision tree and Logistic Regression.

Algorithm:-

1. Load dataset
2. preprocess data
3. apply J48
4. Apply logistic
5. Compare accuracy

code:-

@ relation bank

@ attribute age numeric

@ attribute balance numeric

@ attribute loan {yes, no}

@ attribute default {yes, no}

@ data

25, 1000, no, no

30, 1500, no, no

35, 2000, yes, yes

40, 2500, no, no

45, 3000, yes, yes

50, 3500, no, no

60, 4500, no, no

output:-

Decision Tree (J48)

Correctly classified instance = 5 (62.5%)

Incorrectly classified instance = 3

Kappa statistic = 0

Total instance = 8

Logistic Regression

= 8 (100%)

= 0

= 1

= 8

Result:- Logistic Regression performs better



## Experiment - 4

Aim:- To generate frequent itemsets and rules from book data sets [BOOKS ARFF + Association Rule]

code:-

@relation books

@attribute notaminig {yes,no}

@ attribute DBMS {yes,no}

@ attribute A1 {yes,no}

@ attribute OS {yes,no}

@ data

yes yes yes no

no yes no yes

yes no yes no

no yes yes no

yes yes no yes

no no yes yes

output:- Apriori ?  $L_1=8, L_2=13, L_3=3$

Rules:

OS  $\Rightarrow$  A1

A1  $\Rightarrow$  DBMS

DBMS  $\Rightarrow$  A1

Fp-Growth: No result found (because confidence threshold is too high, so rule satisfied)

Result:-

Apriori and Fp-Growth produce similar rules.

## Experiment - 5

Aim: To compare Decision tree and Rule-based classification

code:-

@ relation credit

@ attributes age numeric

@ attributes income numeric

@ attributes student {yes, no}

@ attributes rating {fair, excellent}

@ attributes class {good, bad}

@ data

25, 1000, no, fair, good

30, 4000, no, excellent, good

35, 5000, yes, fair, bad

40, 6000, no, fair, good

45, 7000, yes, excellent, bad

50, 8000, no, excellent, good

55, 9000, yes, fair, bad

60, 10000, no, fair, good

output:-

Decision tree (J48)

Rule-based (JRip)

Rule-based (PART)

Accuracy = 62.5%

100%

62.5%

correctly classified = 5

8

5

incorrectly classified = 3

0

3

Result:-

JRip classifier performs better than J48, PART

## Experiment-6

Aim: To create a PFF dataset and compare Naïve Bayes and decision Tree Classifiers using accuracy & F1-score.

Code:-

@ relation playtennis

@ attribute outlook {sunny, overcast, rain}

@ attribute temperature {hot, mild, cool}

@ attribute humidity {high, normal}

@ attribute windy {weak, strong}

@ attribute play {yes, no}

@ data

sunny, hot, high, weak, no

sunny, hot, high, strong, no

overcast, hot, high, cool, yes

rain, mild, high, weak, yes

rain, mild, high, weak, yes

rain, cool, normal, strong, yes

overcast, cool, normal, strong, yes

sunny, mild, high, weak, no

sunny, cool, normal, weak, yes

rain, mild, high, weak, no

sunny, mild, normal, strong, yes

overcast, mild, high, strong, yes

rain, mild, high, strong, no.

output:-

Decision tree(J48)

correctly classified instances = 8

Accuracy = 57.14%

incorrectly classified instances = 6



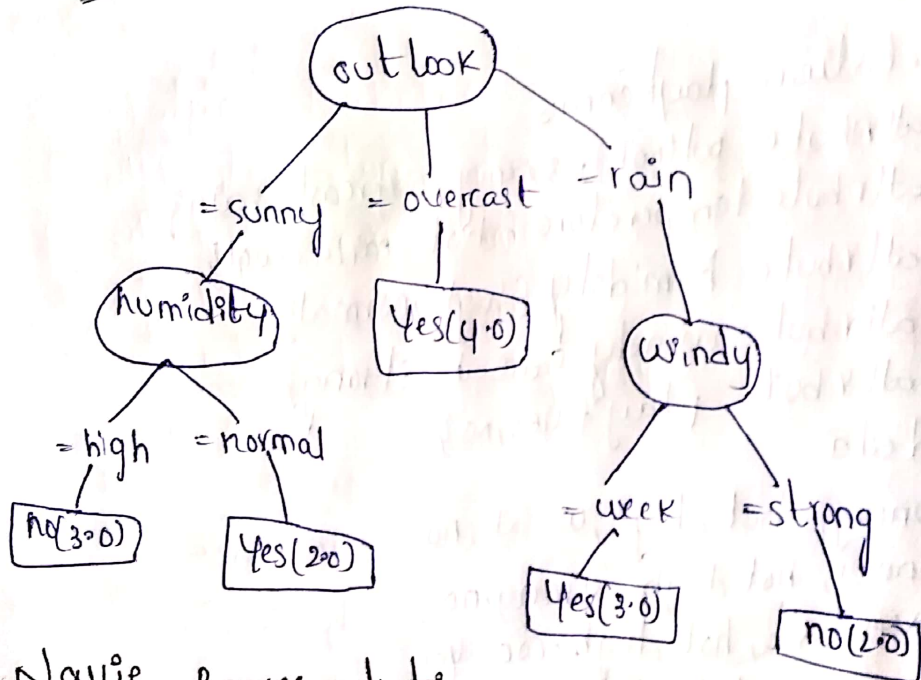
## Confusion matrix

a b

5 4 1 a = yes

3 2 1 b = no

## decision tree:-



## Naive Bayes output:-

correctly classified instances = 7

Accuracy = 50%

Incorrect classified instances = 7

F1-measures = 0.539

## confusion matrix:-

a b

7 2 1 a = yes

4 1 1 b = no

## Result:-

Decision tree classifier performed better than Naive Bayes because it achieved higher accuracy and F1-Score.

## Experiment - 7

Aim:- To perform K-means clustering and visualize employee data on Weka

code:-

Employee Id, Gender, Age, Salary, Credit

111, male, 28, 150000, 39

222, male, 25, 150000, 27

333, female, 26, 160000, 42

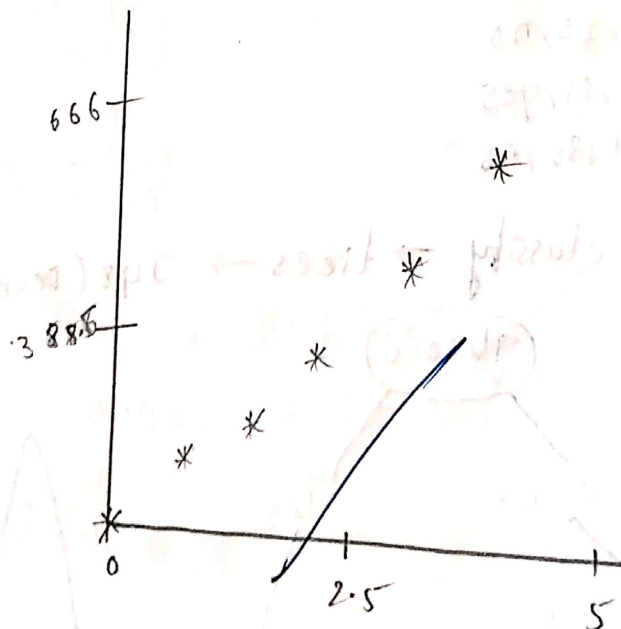
444, female, 25, 160000, 46

555, female, 30, 120000, 64

666, male, 29, 200000, 72

output:-

cluster visualize



Result:-

Employees are grouped successfully based on salary and credit score.

Aim: To predict diabetes using Decision tree and support Vector machine classifiers in weka.

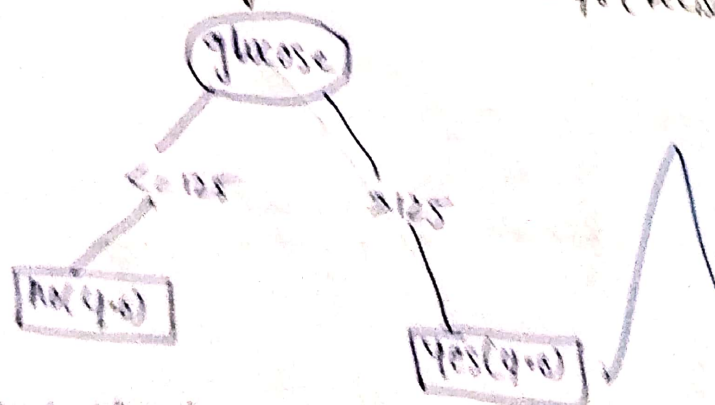
code:

- @ relation diabetes
- @ attribute glucose numeric
- @ attribute bmi numeric
- @ attribute age numeric
- @ attribute diabetes yes/no

@ data

120,35,30,no  
 240,30,45,yes  
 130,28,40,yes  
 110,24,25,no  
 150,35,35,yes  
 100,20,25,no  
 160,38,35,yes  
 125,27,35,no

output: classify → trees → J48 (decision tree)



J48  
 accuracy: 87.5%  
 F1-score: 0.889

accuracy: 100%  
 F1-score: 1.000

Result:

SVM classifier performs better than Decision tree for diabetes prediction



## Experiment - 9

Aim:- To identify the most significant attribute using infoGain evaluator in weka.

code:-

@relation student grades

@attribute marks numeric

@attribute Attendance numeric

@attribute Result {pass, fail}

@data

85, 90, pass

78, 85, pass

92, 45, pass

35, 40, fail

40, 50, fail

88, 92, pass

30, 35, fail

70, 75, pass

60, 65, pass

25, 30, fail

output:-

Ranked attributes

0.931 2 attendance

0.931 1 marks

Selected attributes : 2 1 2

Results:- marks attribute has the better information gain.

## Experiment -10

Aim:- To perform K-means clustering and visualize clusters in weka.

code:-

Employee ID, Gender, Age, salary, Credit

111, male, 28, 150000, 39

222, male, 25, 150000, 29

333, Female, 26, 160000, 42

444, Female, 25, 160000, 40

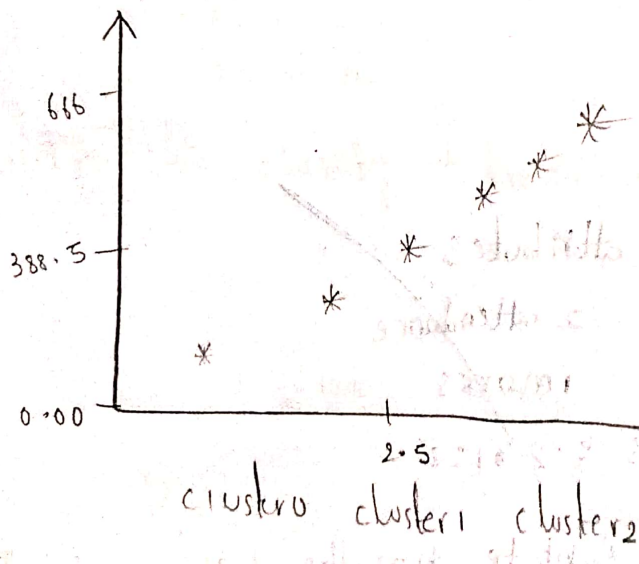
555, Female, 30, 170000, 64

666, male, 29, 200000, 72

output:-

3 clusters formed

High salary and low salary groups identified



Result:- K-means successfully grouped employees into clusters.

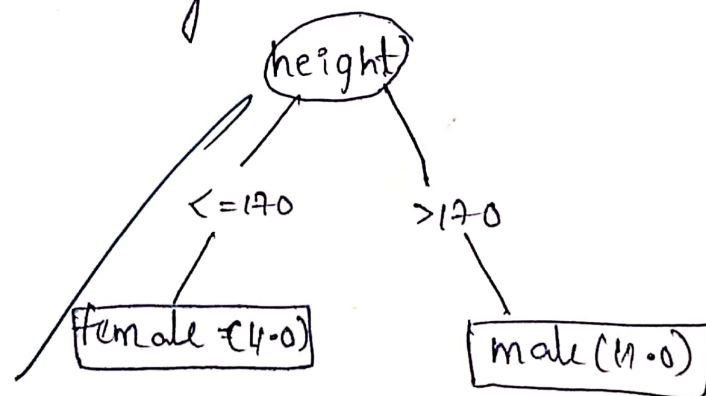
# Experiment - 11

Aim:- To generate rules from a decision tree and compare with rule-based induction

Code:-

@ relation gender  
@ attribute height numeric  
@ attribute weight numeric  
@ attribute gender {male, female}  
@ data  
185, 85, male  
170, 60, female  
182, 90, male  
160, 55, female  
190, 100, male  
165, 58, female  
175, 70, male  
158, 50, female

output:- classify → J48 → start (visualize tree)



J48 accuracy : 87.5%

JRip accuracy = 100%

Result:- JRip generated accurate rule and performed better than J48.